Full measured-boundary FRF phase (Welch H1 reference)

Gray samples select the primary model branch using integer 360-deg shifts. Each earlier model gets at most one constant whole-curve shift.

Display only: reported phase errors use untouched complex FRFs. Independent-unwrap diagnostic is also saved.

Experimental FRF (validation Welch H1) — Before repair — After repair

$u_1 \to y_1$ $u_2 \to y_1$ $u_3 \to y_1$ $u_4 \to y_1$

$u_1 \to y_2$ $u_2 \to y_2$ $u_3 \to y_2$ $u_4 \to y_2$

$u_1 \to y_3$ $u_2 \to y_3$ $u_3 \to y_3$ $u_4 \to y_3$

$u_1 \to y_4$ $u_2 \to y_4$ $u_3 \to y_4$ $u_4 \to y_4$

Unwrapped phase (deg)

Frequency (Hz)